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The Paradoxes of Patenting: Invention and Its Diffusion in 18th- and 19th-Century Britain, France, and North America

C H R I S T I N E M A C L E O D

It is a commonplace of economic analysis that patent systems embody an inevitable paradox: in order to stimulate invention they must inhibit diffusion. Joan Robinson has defined a patent as “a device to prevent the diffusion of new methods before the original inventor has recovered profit adequate to induce the requisite investment.” And, in the words of Nathan Rosenberg, it is “an inescapable aspect” of a patent system “that, to the extent that it offers protection to the holder of a patent, it does so by slowing down the diffusion of inventions.”¹ Yet not all patents are equally inhibitory. Historically there has been much experimentation—both unintentional and deliberate—to establish an optimal balance between invention and diffusion. Much of the debate on the desirability of patent systems, both in the 19th century and among historians today, has centered on the extent to which they promoted invention: would invention occur spontaneously (and be made public) in the absence of a patent system or did it need such a stimulus?² Relatively little attention has been paid

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¹Joan Robinson, *The Accumulation of Capital*, 3d ed. (London, 1969), p. 87; Nathan Rosenberg, *Technology and American Economic Growth* (New York, 1972), p. 188. See also Kenneth Arrow, “Economic Welfare and the Allocation of Resources,” in Nathan Rosenberg, ed., *The Economics of Technological Change* (Harmondsworth, 1971); W. D. Nordhaus, *Invention, Growth, and Welfare: A Theoretical Treatment* (Cambridge, Mass., 1969), pp. 70–91; A. Silberston, “The Patent System,” *Lloyds Bank Review* 84 (1967): 32–44.

²These are discussed in H. I. Dutton, *The Patent System and Inventive Activity during the Industrial Revolution, 1750–1852* (Manchester, 1984), pp. 1–33; Edith T. Penrose, *The*

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to the other side of the equation—the part that 18th- and 19th-century patents played in the commercial exploitation of an invention and the effect of differing configurations of patent laws and administrative practices on the rate and extent of diffusion.³

My first goal in this article is, therefore, to consider the influence of different patent systems on the commercialization and the diffusion of invention. I shall take each of these two related questions in turn, recognizing that there is some artificiality in their separation, especially where a patent was exploited through licensing or assignment. My focus is the unreformed (pre-1852) English patent system, where the emphasis was on invention, to the neglect of commercialization and diffusion.⁴ The French and North American comparisons serve different functions. The former is instructive because it provides a strong contrast with contemporary British practice, demonstrating that it was not inevitable that the 18th-century state should order its priorities in this way. The latter's value lies chiefly in suggesting that the fine-tuning of very similar systems could have unexpectedly large ramifications: that, perhaps counterintuitively, the more certain position of patentees under the United States laws of 1790 and 1793, in comparison with their British counterparts, may have encouraged them to exploit their monopoly powers in ways that were more likely to promote the diffusion of their invention.

Unavoidably, these are very big questions, and in addressing them here I can only hope to raise some of the issues and speculate on the influence of patent laws and their judicial interpretation. It should be understood that I am making a comparison of ideal types, drawn from three countries, two of which tried different patent systems in the course of a century and a half, rather than offering any comprehensive comparison between the technological histories of Britain, France, and the United States. As yet we are not in a position to draw firm conclusions about how the majority of inventors in this period exploited their patents. There has been little research beyond a few, probably untypical, cases—Arkwright, Watt, Leblanc, Blanchard, or

Economics of the International Patent System (Baltimore, 1951), pp. 31–41; and Moureen Coulter, "Property in Ideas: The Patent Question in Mid-Victorian Britain" (Ph.D. diss., Indiana University, 1986).

³For an excellent study of how patent monopolies were exploited by U.S. businesses in the early 20th century, see David F. Noble, *America by Design: Science, Technology, and the Rise of Corporate Capitalism* (New York, 1979), pp. 85–109.

⁴Until 1852, Britain had three separate patent systems: one each for England and Wales, Ireland, and Scotland. Their administrative characteristics were very similar, but the English system was the first and only recourse of the majority of British inventors. I shall use "English" when referring to the patent system, and usually "British" otherwise.

Oliver Evans, for instance—although we are now aware, thanks to the work of Harry Dutton, of “an active trade in invention” conducted by British patentees in the early 19th century. Dutton charted the methods by which inventors, particularly those without manufacturing facilities of their own, were able to exploit their patents through outright sale, licensing arrangements, or partnerships.⁵

The chance of reaching any clear-cut conclusions regarding the impact of differing patent systems on the diffusion of innovation is further hindered by a factor that has hitherto received insufficient attention, and it is one to which I can give only limited space here. It seems probable that the same invention would be diffused at a different rate depending on whether it represented a product invention or a process invention to its patentee. While it is true in one sense that all innovations in production methods constitute process inventions, there is another, strategic sense in which some do not. To take the example of a cotton spinning machine, it would represent a process invention in the hands of a cotton spinner but a product invention in those of an independent machine maker. The former had the option of monopoly use: he might decide to keep the machine for his own factories and, if possible, to prevent his competitors from employing it during the term of the patent—exemplifying thereby the tension between invention and diffusion that is assumed to be inherent in patents. For the latter, however, there was no such contradiction: his profits must come from the sale of his patented machine to as many customers as possible. Even if he decided not to license other machine makers to produce the invention, its diffusion would proceed among his customers; provided demand did not outstrip supply, it should receive no brake from his monopoly position. Neither should it be unduly retarded by the opportunity presented to extract a monopoly price or royalty for the machine, since this would have to be tempered by a calculus of what the market would bear. Although one cannot guarantee the accuracy of such forecasting, any optimistic pricing was likely to be restrained by the difficulty of persuading potential customers to take the risk of innovation, their normal preference being for the status quo and for “no one to make the improvement.”⁶ My second aim then is to raise this objection to the assumption that all patents equally inhibit diffusion (which I have considered in greater depth elsewhere),⁷ principally so that, while the importance of institutional differences in

⁵Dutton (n. 2 above), pp. 122–42.

⁶Robinson (n. 1 above), p. 87.

⁷See n. 82 below.

the invention-diffusion dichotomy is recognized, it should not be made to bear an explanatory weight that it cannot sustain alone.

This article deals with a period of transition for most patent systems in Europe and North America, from instruments of royal prerogative power to specialized and statute-based weapons of capitalist competition. Until 1852 England was still using a patent system established three centuries before, in the reign of Elizabeth I, and statutorily modified only by a clause of exemption in the Statute of Monopolies (1624) and a modest reform in 1835. Like a plant inherited by a reluctant gardener and allowed to grow without pruning or feeding, its evolution was more the product of government indifference than of any economic policy. Increasingly criticized for both inefficiency and injustice, the patenting ancien régime nonetheless survived through to 1852, and in some respects beyond.⁸ By contrast, in 1790–91, as a result of political revolutions, both France and the United States had established patent systems based on positive statute law and embodying the concept of an inventor's natural right to his intellectual property. Moreover, both had since restructured their systems, once in the case of France (1844) and twice in the case of the United States (1793, 1836). In the same half century most other European countries had also enacted new patent laws.⁹ Ironically, the French system after 1791 and the American system between 1793 and 1836 came closer in practice to the English system—in particular, by abandoning formal examination in favor of simple registration.

Commercialization

To begin the comparison by considering the implementation of inventions in the hands of the patentee, one wishes to know what attention the state gave, in the first instance, to moving the patented invention out of the inventor's laboratory, workshop, or indeed brain, and into commercial use. The state might attempt either compulsion—a requirement to implement the invention on pain of the patent's revocation—or assistance, in the form of financial benefits, whether loans, grants, tax exemptions, or protective import tariffs. The British state employed neither. It registered the patent and then withdrew: subsequent disputes about validity were left to the law courts; commercial success rested entirely with the patentee and market forces. If there was no demand that the patent should be worked within a certain time (which doubtless the struggling inventor appreciated), then neither was

⁸For a full account of how the patent system developed over these three centuries, see Christine MacLeod, *Inventing the Industrial Revolution: The English Patent System, 1660–1800* (Cambridge, 1988), chaps. 1–3.

⁹Penrose (n. 2 above), pp. 12–13, 21–31.

there any hope of a helping hand. In this, England was imitated by the United States between 1793 and 1836 but, I shall argue, with significant differences elsewhere in the system that allowed the North American patentee a smoother path.

The contrast with 18th-century France could hardly be greater. Some fifty years ago, Jacques Isoré called attention to the existence of a patent system in France prior to the act of 1791: for more than two centuries before *les brevets d'invention*, there were *les privilèges*, granted by the king and registered by the Parlement de Paris, which might scrutinize, amend, and even occasionally reject them.¹⁰ These customary procedures were codified in a declaration of 1762 that limited the term to fifteen years and required the patent to be worked within one year. Isoré stressed the position of craft guilds as the obstacle that most *privilégiés* were seeking to overcome: in a closed system, where working practices were rigidly controlled, the *privilège* provided a route for the entry of inventions. It has since been shown, by Harold T. Parker, how this dual approach to manufactures was espoused and developed by Colbert: on the one hand, using guilds and *inspecteurs des manufactures* to regulate the quality of workmanship; on the other, giving positive encouragement to entrepreneurs and inventors through governmental assistance of various kinds. Colbert's 18th-century successors, such as Gournay, Turgot, and Necker, may have attempted to liberalize the former arm of his industrial policy, but they endorsed and pursued its latter with little alteration.¹¹

The grant of either national or local monopoly production was but one among several ways in which the inventor might be rewarded and assisted by the French state: he could reasonably expect a lump sum or a pension for life; a guaranteed market, through payment for each machine or tool he made; ennoblement and exemption from taxes; authority to publish his secret in a limited edition at an inflated price; or the title of *manufacture royale* with its attendant privileges.¹² John Kay, for example, frustrated in his attempts to enforce his English patent, emphasized to the French administration in 1749 that he did

¹⁰Jacques Isoré, "De l'existence des brevets d'invention en droit français avant 1791," *Revue historique de droit français et étranger*, ser. 4, 16 (1937): 121. See also Frederik Neumeyer, "A Contribution to the History of Modern Patent Legislation in the United States and France," *Scandinavian Economic History Review* 4 (1956): 139–50.

¹¹Harold T. Parker, *The Bureau of Commerce in 1781 and Its Policies with Respect to French Industry* (Durham, N.C., 1979), pp. 17–38. See also Charles Ballot, *L'introduction du machinisme dans l'industrie française* (Paris, 1923), pp. 10–11, 17; Pierre Deyon and Philippe Guignet, "The Royal Manufactures and Economic and Technical Progress in France before the Industrial Revolution," *Journal of European Economic History* 9 (1980): 616–26.

¹²Parker (n. 11 above), pp. 47–59.

not want a *privilège exclusif* for his flying shuttle; instead, for instructing weavers in Languedoc in how to use the shuttle, he and his partner received 15,000 livres (£650 approximately); similar sums and annual pensions were awarded him in other cloth-making regions.¹³ In England, the Swiss inventor Aimé Argand found his patent for lamps infringed, and then revoked by a court, within a year of its enrollment in March 1784; in France, authorized to establish a manufacture, he was awarded 24,000 livres (£1,000) to subsidize production. This business appears to have flourished, the lamps (although expensive) selling well, until the Revolution cut short his success.¹⁴

The second major point of contrast between the French and the English systems was the degree of scrutiny to which French applications were subjected; this followed from the respective types of recognition available. While the English attorney general normally took the applicant's word that his invention was his own and novel, the French bureaucracy was put into full operation, in both the center and the locality. Liliane Pérez has demonstrated how the inventor was often first brought to the attention of the Bureau du Commerce by the *intendant* or *inspecteur des manufactures* of his province; he might already have had the invention appraised by local craftsmen and a provincial *académie*.¹⁵ One of the four *intendants* of the bureau would then marshal further evidence. Since 1699 the Académie des Sciences had been required to examine the technical merits of all machines, products, and processes submitted to it, and increasingly the bureau appointed individual scientists as salaried consultants. A verdict would also be sought from appropriate government departments on such matters as the availability of sufficient fuel and transport facilities in the locality, the impact on revenue, the effect on regional or national prosperity, and the circumstances and capacities of the inventor. The *intendant* would present the case before his three colleagues and, if they agreed, send it to the Contrôleur Général for his final decision on what, if any, assistance or reward was appropriate.¹⁶ Of course, the judgment might be negative: there was considerable scope here to find reasons why the invention should not receive

¹³Archives Nationales, F12/993. Kay had vainly tried to secure a monetary reward from the British Parliament: A. P. Wadsworth and Julia De Lacy Mann, *The Cotton Trade and Industrial Lancashire, 1600–1870* (Manchester, 1965), p. 458.

¹⁴Archives Nationales, F12/992; T. Heyer, *Ami Argand, inventeur des lampes à courant d'air* (Geneva, 1861), pp. 21–38.

¹⁵Liliane Pérez, "Invention, politique et société en France dans la deuxième moitié du XVIII^e siècle," *Revue d'histoire moderne et contemporaine* 37 (1990): 39–46, 53–58.

¹⁶Parker (n. 11 above), pp. 4–15, 91–105.

public support. The point is, however, that innovation, as well as invention *tout court*, was seriously addressed.

Essentially, we are looking here at two quite different institutions, in both means and ends, that shared little more than the aim of recognizing and prompting invention. The English patentee paid approximately £100 (£350 to include Scotland and Ireland) for the right to exercise a fourteen-year monopoly over the production and sale of his invention—a monopoly of dubious strength, as will be shown below—and was left to exercise his entrepreneurial skills in the realm of market forces. Originally designed to promote the immigration of Continental craftsmen, the 16th-century English “patent system” had shared many characteristics with that of 18th-century France, both of them steered by government ministers actively concerned to promote the implementation of superior techniques.¹⁷ But in the aftermath of abuse and consequent implication in the mid-17th-century constitutional crisis, it had shriveled by the early 18th century into little more than a minor source of government income, kept afloat by the demand from a small but growing band of inventors and tradesmen. The government’s chief concern was to avoid embarrassment by any suggestion of monopolies or constitutional irregularity. Administrative procedures were accordingly devised to shift responsibility for policing the system onto the judiciary, which began to establish a case law within a minimal statutory framework.¹⁸ Patents were justified by the largely implicit assumption that invention should be rewarded and encouraged. Whether they achieved this goal was rarely questioned before the 19th century.¹⁹

In contrast, the French *privilégié* paid nothing (though the expense incurred in demonstrating his invention to the bureaucracy might be large), and he could hope for a substantial reward. He might be helped to start commercial production, or he might be relieved entirely of that burden for which many inventors were not suited. He could expect careful scrutiny, but he might also receive assistance in presenting his case. (Pérez has found a much higher proportion of provincial patentees in France than in England, which may reflect their easier access, via the local bureaucracy, to Paris; their English counterparts had to make personal application to London.)²⁰ The French government dealt with invention as an integral part of its economic policy, but it did not stop with invention: its concern was to

¹⁷G. D. Duncan, “Monopolies under Elizabeth I” (Ph.D. diss., University of Cambridge, 1976), pp. 36–49.

¹⁸MacLeod (n. 8 above), pp. 20–39, 46–55, 58–64.

¹⁹Ibid., pp. 182–200.

²⁰Pérez (n. 15 above), pp. 49–50.

establish manufactures, and it was prepared to go to considerable lengths to nurture them. One tends to think only of the dead hand of mercantilist regulation, but it ran in parallel with a policy for innovation.

Aspects of both approaches are to be found in the practices of American colonial and then, briefly, state governments prior to the federal act of 1790. Most colonial laws resembled the English Statute of Monopolies and evinced a similar anxiety to limit the excesses of monopolists yet at the same time to encourage the introduction of new manufactures from overseas and promote invention at home. Exclusive monopoly powers were the normal reward, but Virginia granted small sums of money to several inventors.²¹ After the Revolution, inventors such as John Fitch, James Rumsey, and Oliver Evans obtained patents from several states. Further, they expected financial assistance: Fitch petitioned Congress (without success) to purchase 4,000 of the maps he had produced; Rumsey was granted 30,000 acres of land west of the Ohio, but lost it again when he failed to meet the condition Congress had imposed—a working prototype within one year. He did, however, secure the informal support of the “American establishment,” which subscribed money and opened doors across Europe, including that of Matthew Boulton.²²

Before running away with too glowing a picture of the inventor's lot in either France or North America, it is worth taking cognizance of the negative aspects of close state involvement. Washington, Franklin, and Jefferson threw their money or prestige behind Rumsey; his rival steamboat pioneer, Fitch, was frozen out, failed to get financial support, and was badly hindered by his lack of funds. The introduction of the new patent law in 1793 seems to have served only to deepen the stalemate of their antagonism.²³ In France, as Charles Gillispie has demonstrated, an inventor could be ruined by an official whose lack of faith in either the technical or commercial merits of his proposal or whose personal animosity could prevent him even attempting its commercial exploitation. He might also suffer from a conflict between government departments or between the center and the locality or from the territorial restrictions imposed by rival *privilégiés*. It was far from plain sailing. Such incidents were instrumental in the National Assembly's rejection of examination, in favor

²¹Floyd L. Vaughan, *The United States Patent System: Legal and Economic Conflicts in American Patent History*, 2d ed. (Westport, Conn., 1972), pp. 16–18; Neumeyer (n. 10 above), pp. 127–38.

²²Brooke Hindle, *Emulation and Invention* (New York, 1981), pp. 31–43.

²³*Ibid.*

of simple registration, in the 1791 law.²⁴ At least in England or America one could risk one's own money on any apparently crazy scheme, patented or unpatented; one did not need the state's permission to start a manufacture.

Witnesses before 19th-century British parliamentary select committees debated furiously whether worthwhile inventions would always find financial backing. The debate continues, and Harry Dutton recently argued that "there is scant evidence to support Landes's view that inventors had little difficulty in finding funds for development."²⁵ Much depended on the circumstances of the individual inventor and the nature of his invention: for instance, the more radical a break with current practice it implied, the more difficult it was for him to interest manufacturers in it and generally the more finance and time he needed to develop it. This problem arose independently of the patent system, of course, but the high cost of British patents compounded it. The necessity to seek funds simply to patent an invention put the inventor in a far weaker bargaining position than if he had been able to approach a capitalist, patent in hand; simultaneously, he ran a substantial risk of losing his unprotected secret in the course of negotiations. There is repeated reference in the 1851 select committee's evidence to the vulnerability of inventors in this regard; particularly at risk was the inventive workman, who "has not generally the means of making terms with his master as to invention."²⁶ Investment in an American patent, costing \$30 (equivalent to approximately £7), or a French one, which went from 300 livres (approximately £13) for five years to 1,500 livres for fifteen, under their respective new laws, required much less deliberation. It is worth noting Kenneth Sokoloff's remark that even \$30 represented "a rather substantial fee" and a "non-trivial cost."²⁷ The high cost and the restriction of shareholders in a patent to five (twelve after 1832) were the only criticisms made of the English system by a French report to the National Assembly in 1790 that otherwise preferred the "laissez-faire" that operated across the Channel to the state's involvement at home.²⁸

²⁴Charles C. Gillispie, *Science and Polity in France at the End of the Old Regime* (Princeton, N.J., 1981), pp. 463–78; John Graham Smith, *The Origins and Early Development of the Heavy Chemical Industry in France* (Oxford, 1979), p. 202; see also Pérez (n. 15 above), pp. 48–49.

²⁵Dutton (n. 2 above), p. 169.

²⁶*Report from the Select Committee of the House of Lords [on Patent Law Reform]*, British Parliamentary Papers (hereafter BPP) 1851, XVIII, p. 254; see also pp. 319, 422–23, 447, 483, 522, 688; and University College, London, Brougham MSS 17, 272.

²⁷Kenneth L. Sokoloff, "Inventive Activity in Early Industrial America: Evidence from Patent Records, 1790–1846," *Journal of Economic History* 48 (1988): 818.

²⁸S.-J. De Boufflers, *Rapport fait à l'Assemblée Nationale . . . sur la propriété des auteurs de nouvelles decouvertes et inventions en tout genre d'industries* (Paris, 1791), pp. 9–19.

Finally, on the question of the state's support for innovation, we may note the success of direct state sponsorship in France. Gillispie emphasizes the importance of initiatives taken by the Bureau du Commerce and other agencies of the state in stimulating major inventions such as "Vaucanson's looms, Holker's cotton machinery, coke-fired smelting at Le Creusot, the Montgolfier hollanders." These inventions resulted from the identification of a need or problem that had led to the establishment of a research program or a prize to promote the discovery of a solution.²⁹ In England, a few monetary awards were made by Parliament but they were post facto, usually to compensate a valued inventor whom the patent system had failed (for instance, Samuel Crompton or Edmund Cartwright).³⁰ The one exception was the offer of prizes for the means of establishing longitude: in response to it and financed by awards made for his early efforts, John Harrison perfected his chronometer.³¹

Diffusion

If we turn now to look at how far these patent systems respectively promoted or inhibited the dissemination of an innovation beyond its inventor, we find that early-modern governments (if not some of their successors) already knew what historians have been rediscovering about the importance of personal demonstration in the transfer of technology.³² Part of the point of requiring the invention to be put into use within a certain time was that others, not least the patentee's employees, should discover his methods. France introduced a "working" clause in 1762 and retained it in the new law of 1791. This could

²⁹Gillispie (n. 24 above), p. 463. Gillispie is surely too skeptical in doubting that the "grass-roots inventions" brought to the Bureau du Commerce ever entered production: Deyon and Guignet (n. 11 above) and the current research findings of Liliane Pérez suggest they frequently did. One clear example of grass-roots initiatives' being more successful than those of competitors for a prize is in the manufacture of soda from salt: Smith (n. 24 above), pp. 198–206.

³⁰Klaus Boehm and Aubrey Silberston, *The British Patent System: I, Administration* (Cambridge, 1967), pp. 25–26.

³¹Humphrey Quill, *John Harrison, the Man Who Found Longitude* (New York, 1966), pp. 35–37, 40–48, 68–69. It proved almost as great a challenge to secure his full reward from a parsimonious government as it had been to find the longitude.

³²J. R. Harris, "Skills, Coal and British Industry in the Eighteenth Century," *History* 61 (1976): 167–82, and "Attempts to Transfer English Steel Techniques to France in the Eighteenth Century," in Sheila Marriner, ed., *Business and Businessmen: Studies in Business, Economic, and Accounting History* (Liverpool, 1978), pp. 199–233; Peter Mathias, "Skills and the Diffusion of Innovation from Britain in the Eighteenth Century," *Transactions of the Royal Historical Society*, 5th ser. 27 (1977): 90–113; David J. Jeremy, *Transatlantic Industrial Revolution: The Diffusion of Textile Technologies between Britain and America, 1790–1830s* (Cambridge, Mass., 1981), pp. 20–33, 76–79.

be supplemented by the requirement to instruct a certain number of apprentices—a clause also found in English patents regularly in the 16th century, occasionally in the 17th, but never in the 18th³³; it was still to be found, however, in some American colonial patents. It was associated principally with the importation of entire new manufactures from overseas. We have also seen, in the case of prerevolutionary France, how an inventor might be paid a lump sum or a pension to teach his methods or be granted copyright in a limited edition where he would reveal his secret. The French were also actively securing the person of the inventor by giving him encouragement to establish a manufacture. United States patents were restricted to citizens and aliens resident for at least two years, but, in contrast with France, imported inventions were disallowed. There was no requirement for the holder of an English patent either to reside in England or to work his patent. These liberal conditions helped make England a magnet for inventions, but it is unclear how many of them were commercialized.³⁴ The concern over unworked patents, particularly those held by nonresidents, that was to lead to provision for compulsory licenses in the 1883 act and the “compulsory working” clause in the 1907 act was already being voiced in 1851.³⁵

It was the specification, a detailed description of the invention submitted several months after the patent had been sealed, that everywhere came to carry the burden of securing the inventor's secret and informing the public. By the early 19th century it was a common theoretical justification, offered by both British and North American writers, to regard a patent as a contract struck between the public and the inventor, who was awarded a temporary monopoly in return for disclosure of his secret.³⁶ Considerable weight was correspondingly attached to a clear, full, and accurate specification, sufficient in itself to allow one skilled in the relevant trade to imitate his methods. Both the idea of a contract and of this role for the specification were, however, slow to emerge. Specification was regularly required of English patentees after 1734, but, as with the original application, the specification was not scrutinized unless and until the patent came

³³A “compulsory working” clause was introduced in the 1907 act to permit the revocation of British patents being worked overseas but not in Britain within four years of the grant; A. F. Ravenshear, *The Industrial and Commercial Influence of the English Patent System* (London, 1908), pp. 82–88.

³⁴BPP 1851 (n. 26 above), XVIII, pp. 335–36, 392–93, 478, 489; De Boufflers (n. 28 above), pp. 20–21.

³⁵BPP 1851, XVIII, p. 391; Boehm and Silberston (n. 30 above), p. 30; see also n. 33 above.

³⁶Dutton (n. 2 above), p. 22.

before the law courts. There was no “quality control” over its accuracy other than this ultimate threat. Nor was specification originally *intended* as a means of informing the public; it was, rather, a measure introduced to assist the law courts in deciding the merits of disputed patents since the executive had declined to do so.³⁷ This was the corollary of a rationale for the patent system that stopped short—at invention. Most commentators regarded patents as perquisites to reward inventors and encourage others to emulate them; dissemination of that invention was hardly considered. It was only in the final quarter of the 18th century that the English judiciary began to make the specification the focal point of the patent system and to formulate the notion that it should inform the public. This was adopted in the U.S. federal law in 1790, and in France formally in 1791, although there was explicit provision to keep specifications secret (from foreigners in the first instance).

Specification became the patentee’s Achilles’ heel at law. Dutton has shown that until 1830 English judges and juries were generally biased against patentees and lost few opportunities to find the specification deficient.³⁸ Even when this prejudice dissolved, legal costs and slow procedures remained a problem. Bennet Woodcroft testified in 1851 that he had spent £3,000 in prosecuting two licensees under his patent for loom tappets who had “found them beneficial, and then they made them for themselves”; 300,000 patent tappets had since been made, but “I am still a loser from it.” This was quite a modest sum by contemporary levels of legal fees in patent cases.³⁹ Patentees were accordingly reluctant to go to court and often, after threats of prosecution had had no impact, tolerated infringements for years or compromised with the chief offenders. The patent agent and engineer John Farey told the House of Lords inquiry in 1835 that, despite “the Courts exercising their discretion in a much more favourable manner towards patentees than they had done formerly,” the law still needed to be changed because “there is no alteration in the feeling of insecurity and want of confidence of protection by the law. . . . [Inventors] feel discouraged, instead of encouraged, to proceed.”⁴⁰ As a result principally of this insecurity, those patentees who were in a position to do so, for example manufacturers using their own

³⁷MacLeod (n. 8 above), pp. 48–55.

³⁸Dutton, pp. 76–81.

³⁹British Patent no. 7,529 (1838); BPP 1851, XVIII, p. 477. For other examples, see Dutton (n. 2 above), pp. 181–82.

⁴⁰House of Lords Record Office, Minutes of Evidence on Patents Amendments Bill (hereafter HLRO, Minutes), July 6, 1835, pt. II, f. 8; see also ff. 21–22; and, e.g., BPP 1851, XVIII, p. 516.

invention, might attempt to work their patent in secret. This may appear paradoxical: surely the specification was supposed to inform the public; surely the choice was *between* a patent and secrecy.

Theory and practice were, in fact, poles apart. Informing the public simply meant securing a written description for posterity, not disseminating it. Specifications were released in a haphazard manner. English patents were announced officially and briefly in the *London Gazette*; specifications were not published and were enrolled in any of three separate offices, randomly. If one knew the name of the patentee, one could pay a fee to have the clerk at each office in turn search for it and purchase a copy at a cost (by 1829) of between 2 and 40 guineas (depending on its length and the number of drawings).⁴¹ One could not ask for copies of all specifications concerning a particular technology (though one could enter a caveat to be informed of all such applications): the request had to be by name. And, for the sake of greater anonymity, a manufacturer with a reputation for successful innovation would sometimes patent in the name of his foreman or other trusted employee.⁴² From 1794, *The Repertory of Arts and Manufactures* attempted to alleviate this situation by publishing what it considered to be the most important specifications, but there were constraints of both space and cost. David Jeremy estimates that only a quarter of specifications were ever published, including a minority of those that proved most worthwhile. He further suggests how poor an instrument of industrial espionage patents were.⁴³ Many patentees feared otherwise: a systematic surveillance (through caveats and regular inquiries) could provide the basis for both direct imitation and a prepared challenge to any patent's validity; perusal of a specification might permit infringements, whose prosecution would be too expensive to pursue.⁴⁴

A parallel problem existed in the United States. Again it manifests the common assumption that patenting and diffusing an invention were necessarily antithetical—the assumption that lay at the heart of the pejorative identification of patents with monopolies. The first superintendent of the Patent Office, William Thornton, respecting the wishes of patentees who doubted the security of their patent, first contested and then reluctantly followed the attorney general's instruction to reveal the specification of unexpired patents to individual

⁴¹Jeremy (n. 32 above), p. 47.

⁴²For example, William Radcliffe, *Origins of the New System of Manufacture* (Stockport, 1828), p. 26; Manchester Central Library, Mason MSS, Box 2, f. 55.

⁴³Jeremy (n. 32 above), pp. 46–48.

⁴⁴Dutton (n. 2 above), pp. 184–85; Eric Robinson, "James Watt and the Law of Patents," *Technology and Culture* 13 (1972): 118–35.

inquirers. But when the secretary of the Franklin Institute asked him to supply recent specifications for publication in its new journal, in 1825, Thornton dug in his heels. After a fierce legal dispute, he was required to comply. Thornton's view was that ideally the specifications should remain confidential until the term had expired; otherwise, inventors might decide not to risk patenting and their inventions consequently would never be secured by the public.⁴⁵ On the other hand, effective confidentiality posed a major problem to potential patentees attempting to discover what was already subject to a patent in their sphere. This was urged on the 1851 British parliamentary select committee by Bennet Woodcroft as a reason both for publishing full specifications and regular indexes to patents and for making them widely available in libraries.⁴⁶ Under normal circumstances before 1852 the specification informed few, but it provided one more channel for those who resorted to industrial espionage by whatever means were available.

It was ultimately the expense and uncertainty of litigation, whether as a plaintiff or defendant, that drove many English patentees to attempt secrecy. They dared not risk preemption and exclusion from use of their own invention (a patent at least assured that much); yet neither did they dare risk litigation to crack down on infringements. The answer for some was to purchase a patent as quickly and anonymously as possible, in order to establish priority; enter a specification, perhaps attempting in some subtle way to baffle anyone who tried to use it (at little additional risk if litigation did ensue, since before the 1830s even the most explicit specification was liable to be found faulty by the courts); swear the workmen to secrecy, prohibit any stranger from entering the works, and proceed to make and use the invention.⁴⁷ Industrial espionage on site or the defection of workmen was much more likely to spring the secret than was anyone searching for the specification in Whitehall. These secretive patentees constituted a category additional to that of inventors who attempted secrecy *instead of* a patent because they considered its protection either useless or too expensive (concerning whose numbers contemporaries disagreed sharply—an unresolved puzzle bequeathed to historians).⁴⁸

⁴⁵Daniel Preston, "The Administration and Reform of the U.S. Patent Office, 1790–1836," *Journal of the Early Republic* 5 (1985): 339–42.

⁴⁶BPP 1851, XVIII, pp. 460–61; also Ravenshear (n. 33 above), p. 96.

⁴⁷For examples, see *Report from the Select Committee . . . Law and Practice Relative to the Granting of Patents for Invention*, BPP 1829, III, pp. 502–4.

⁴⁸*Ibid.*, pp. 548–51; BPP 1851 (n. 26 above), XVIII, pp. 423, 513; *Reports from the Assistant Hand-Loom Weavers Commission*, BPP 1840, XXIII, p. 240; *Quarterly Review* 43 (1830): 336; HLRO, Minutes, July 6, 1835, pt. I, ff. 3, 9.

It is now time to introduce the distinction made earlier between product and process inventions: the same machine or tool could be either, depending on the business of its producer. Secrecy was a viable strategy only for process inventions; products, by definition, had to be made public and open to inspection. Where production machinery was concerned it followed that the option of secrecy was restricted to inventors in a position to profit simply by using it. For machine makers, sale, and therefore diffusion, was the only course.

These different strategies are clearly visible in an example from the paper industry. Of two papermaking machines patented in England in the first decade of the 19th century, one became standard there, the other in the United States. The former was the Fourdrinier machine, produced by the engineer Bryan Donkin in London for sale to papermakers. By 1851 Donkin's firm had made 191 machines, including 108 for export; they were also produced by several other firms once the patent expired in 1821.⁴⁹ Donkin's policy was to patent his improvements and to place no restrictions on anyone's viewing his products. (His machine tools were, however, kept secret, neither patented nor for sale.) The latter was the cylinder-mold machine, invented by John Dickinson for his own use and patented in 1809. Dickinson sought his fortune not in selling the machines but only in using them to make paper. His policy differed accordingly from Donkin's: he was notoriously secretive, prohibiting strangers from his mills and making the machinery on site. His patent specifications, drawn up by himself, were reported in 1835 to be insufficient "to enable others to attain the same results, notwithstanding the information given." Dickinson's machinery was sufficiently different from the Fourdrinier to merit an uncontested patent, but equally viable; it was, however, deliberately limited to his own paper mills.⁵⁰

Ironically, the Dickinson machine became more common than the Fourdrinier in the United States thanks to a classic piece of transatlantic industrial espionage that ensured its transference eleven years before Donkin exported two of his machines there in 1827. Simplified in its construction by several American papermakers in the 1820s, the

⁴⁹R. H. Clapperton, *The Paper-making Machine: Its Invention, Evolution and Development* (Oxford, 1969), pp. 16–33, 43–63, 225–26; Alistair G. Thomson, *The Paper Industry in Scotland, 1590–1861* (Edinburgh, 1974), pp. 169–78; *Exhibition of the Works of Industry of All Nations, 1851, Reports of the Juries* (London, 1852), 17:427.

⁵⁰British Patent no. 3,191 (1809), no. 3,452 (1811), no. 3,839 (1814); Eugene S. Ferguson, ed., *Early Engineering Reminiscences (1815–40) of George Escot Sellers* (Washington, D.C., 1965), pp. 118–30; Joan Evans, *The Endless Web: John Dickinson & Co. Ltd, 1804–1954* (London, 1955), pp. 8–22; HLRO, Minutes, July 6, 1835, pt. II, f. 10; Clapperton (n. 49 above), pp. 65–70; Thomson (n. 49 above), pp. 158, 193 n. 5.

cylinder-mold machine enjoyed the advantages of a lower capital cost and of the familiarity that sprang from its head start. The Fourdrinier machine found its main market in areas where mechanized papermaking began relatively late, and among makers of superior quality paper, for whom its ability to produce paper of equal strength in both directions justified the higher initial cost. American machinists offered both types of machine to their customers, whose choice could be exercised in the light of their own requirements and capabilities.⁵¹ In England, Dickinson enjoyed a reputation for high-quality products. This may have resulted from his machine's ability to produce both the fashionable thin papers and watermarked paper earlier than the Fourdrinier could, or it may have stemmed from the benefits of more knowledgeable maintenance and operation that accompanied in-house machine building. Whatever the cause, it suggests that some British papermakers should have had an incentive to emulate the patentee and to prefer the cylinder-mold machine, had they been offered the choice. If Dickinson's machine suffered the reputation of being more complicated, as was reported in 1829, wider exposure to more mechanical talents might have permitted the sort of simplifying improvements that were being made to it in the United States.⁵²

Dickinson made his fortune by producing paper, not by exploiting a patent. In what other ways, defensive or offensive, he might have managed his patent if the Fourdrinier machine had not been available to rival papermakers is a matter for speculation. But he put his faith ultimately in maintaining secrecy, not in patent protection. Knowledge of his machinery was transmitted to the United States not via his specification but by one of his workmen. Since, like most English patentees at this time, he had no American patent, he was unable to prevent this, and in any case it offered no direct competition. The Fourdriniers, in contrast, were bankrupted by their investment in developing their machine and the initial reluctance of papermakers to buy it. (One may speculate how much better they or the machine's French inventor might have fared in France before 1791.) Donkin survived and modestly prospered by working the patent openly: as a machine maker, secrecy was not an option for him; competition was deterred less by the patent than by the capital investment required

⁵¹In 1882 61 percent of all U.S. papermaking machines were of the cylinder-mold type: Judith A. McGaw, *Most Wonderful Machine: Mechanization and Social Change in Berkshire Paper Making, 1801–1885* (Princeton, N.J., 1987), pp. 101–2, 158–83.

⁵²Evans (n. 50 above), pp. 11, 21; *Report from the Select Committee on Fourdrinier's Patent*, BPP 1837, XX, p. 70. My thanks to Judith A. McGaw for her invaluable elucidation of papermaking technology.

and the initially poor market for machines.⁵³ It was in his interest that the machine should be diffused as widely as possible under his aegis, and the royalty payable to the Fourdriniers seems to have added only a small deterrent to potential purchasers on top of the much greater capital cost.

Such tactics as Dickinson's were, of course, implausible if one aimed to exploit the invention under licensing agreements or in any way that had to involve other parties. The dilemma was most acute for manufacturers whose patent covered a process invention. Like Dickinson, the patentee's preference might be to monopolize its use in order to gain a competitive edge in his own industry, but, if the invention were of any importance (and there were no obvious alternatives), this strategy would place him under enormous pressure from rivals seeking to share the benefits either licitly or illicitly. Licensing was a way of defusing opposition and, at the same time, securing an additional gain through royalty payments. But setting the level of royalties involved some nice judgments. It is hardly surprising that patentees sometimes erred into provoking infringement by setting it too high—and other patentees reacted by withdrawing into secrecy. Some of the more famous inventors of the late 18th century illustrate these difficulties. Of the "textile triumvirate" of Hargreaves, Crompton, and Arkwright, only the last named was financially successful. Samuel Crompton made no attempt to patent the spinning mule and failed to keep its secret for himself.⁵⁴ James Hargreaves's patent for the spinning jenny was rapidly and massively infringed, Hargreaves apparently being advised that his patent was vulnerable to the charge of prior publication (he had made jennies for sale before obtaining the patent). According to his near contemporary, Gravenor Henson, Hargreaves "shrunk from the expensive contest" and suffered obscurity and distress. Any money he made was from his own spinning partnership.⁵⁵

Richard Arkwright's case is the most instructive. He appears initially to have had no intention of licensing others to use the water frame or the associated preparatory machinery.⁵⁶ His *modus operandi*

⁵³BPP 1837, XX, pp. 53, 54, 56.

⁵⁴George W. Daniels, *The Early English Cotton Industry* (Manchester, 1920), pp. 116–18, 168.

⁵⁵British Patent no. 962 (1770); Gravenor Henson, *The Civil, Political and Mechanical History of the Framework Knitters, in Europe and America*, ed. S. D. Chapman (Newton Abbot, 1970), p. 366; Wadsworth and Mann (n. 13 above), pp. 477–82; Christopher Aspin and Stanley Chapman, *James Hargreaves and the Spinning Jenny* (Helmshore, 1964), pp. 26, 38–39.

⁵⁶British Patent no. 931 (1769), no. 1,111 (1775).

was to establish a series of spinning mills through partnerships with capitalists who wished to emulate his success. In a letter of 1778, inquiring about the legal niceties of issuing licenses to third parties, he stated that he had “never yet sold any” (nine years after patenting the water frame). It is not known how many licenses Arkwright subsequently issued, but the price he charged for them—variously estimated at £2–£7 per spindle—seems to have deterred all but the most “law-abiding.” Other manufacturers chose to risk litigation, and, when Arkwright moved to prosecute their infringements, they banded together to defeat his patents. Four years of ultimately unsuccessful litigation cost Arkwright over £2,200.⁵⁷ When he did secure a verdict against one offender, he was advised that “as it seems the injuries were committed very much in secret there is no other way . . . to obtain an adequate satisfaction. I rather think that a separate bill must be filed against each invader of the patent and this will make it necessary for the patentees to consider whether any injury they can have sustained just before the expiry of their patent can make it worth while to enter upon so extensive a field of litigation.”⁵⁸

Nonetheless, Arkwright died a very wealthy man. His profits came from spinning cotton, not from patent royalties. Arguably the patent was essential for him to secure the advantage he made from the former, but it was notoriously a period when large fortunes were made in cotton spinning by men who shared Arkwright’s business acumen, if not his patent protection. It was Matthew Boulton’s opinion that Arkwright would have kept his patents and profited by licensing others to use them if only he had set the royalty at a more realistic level.⁵⁹ Given his initial reluctance to license, one may wonder whether Arkwright set the level too high not because he misjudged his market but because he was still attempting to restrict diffusion and capitalize on his monopoly. There seem to have been clauses in his license agreements to stint the output of his direct competitors through penalties for “overspinning.”⁶⁰

Arkwright’s problems were closely monitored by James Watt, who appeared twice as a witness in defense of Arkwright’s patents. Watt felt beleaguered. The successive defeats of Arkwright and Argand made him believe his own patents were similarly threatened, and he

⁵⁷R. S. Fitton, *The Arkwrights: Spinners of Fortune* (Manchester, 1989), pp. 75, 91–138, 225; S. D. Chapman, *The Early Factory Masters* (Newton Abbot, 1967), pp. 73–76, 128.

⁵⁸Fitton, *The Arkwrights*, pp. 104–5.

⁵⁹Wadsworth and Mann (n. 13 above), p. 490; Fitton, *The Arkwrights*, pp. 98, 138–40, 182.

⁶⁰R. S. Fitton and A. P. Wadsworth, *The Strutts and the Arkwrights, 1758–1830* (Manchester, 1964), p. 94.

devoted much thought to reforms that would increase the security of patentees. These included a provision to keep specifications secret until the patent expired.⁶¹ It probably also influenced his practice. For a long time he tolerated infringements of the steam engine patent rather than risk its loss in court. He also refused to issue any licenses to other engine makers. Many of the “pirates” who infringed the patent might have been willing to produce and erect engines under license, had that been an option available to them. It seems unlikely that the diffusion of the Watt engine was seriously retarded by his monopoly, not least because Boulton and Watt’s business was to supply their patented product, but also thanks to the “back door” opened by the implications of judicial uncertainty and the stampede of engineers through it—until finally Boulton and Watt virtually closed it for the remaining few years of their (extended) patent by securing some unlikely victories in the law courts in the 1790s. Many of the “pirate” engines might have been better constructed, however, if their builders had had the benefit of Watt’s advice, and the design of the engine would almost certainly have progressed further if Watt had felt able to pursue a more liberal policy.⁶²

The first half of the 19th century saw British patentees learning how to exploit licensing more profitably, if at the cost of extensive litigation. It remained a hazardous business and only a tiny minority mastered the art. One who fumbled his way through reversals of legal fortune to success was John Heathcoat, inventor of the bobbinet machine for lace, first patented in 1808. By 1816 Heathcoat and his partner, Charles Lacy, had each taken profits of between £40,000 and £50,000, principally from their factory of fifty-five frames at Loughborough (Leicestershire) but also through the issue of an unknown number of licenses. The industry’s first historian, William Felkin, remarked on “the dislike felt in Nottingham to the payment of tribute under the patent, which though not exorbitant, was very profitable to the patentees.”⁶³ Three years earlier they had effectively lost the patent in the course of prosecuting an infringement, when it was discovered that a copyist had omitted a line from the specification,

⁶¹Fitton, *The Arkwrights*, pp. 138–40; Robinson (n. 44 above), pp. 118–36.

⁶²Richard L. Hills, *Power in the Industrial Revolution* (Manchester, 1970), pp. 152, 185–86; Jennifer Tann, “Mr. Hornblower and His Crew: Watt Engine Pirates at the End of the 18th Century,” *Transactions of the Newcomen Society* 51 (1979–80): 95–105, and “Boulton and Watt’s Organisation of Steam Engine Production before the Opening of Soho Foundry,” *Transactions of the Newcomen Society* 49 (1977–78): 41–52.

⁶³Stanley D. Chapman, ed., *Felkin’s History of the Machine-wrought Hosiery and Lace Manufactures* (Newton Abbot, 1967), p. 206; British Patent no. 3,151 (1808), no. 3,216 (1809).

and Heathcoat was advised not to proceed. By the time his patent was secured through the judgment in an independent dispute (*Bovill v. Moore*, 1816) that the plaintiff's machine was an infringement of Heathcoat's patent, there were alleged to be 156 unlicensed makers and owners of the bobbinet frames. One successful action in the Court of Common Pleas proved to be sufficient to bring the majority to a compromise: licenses for nearly 700 frames were issued, which produced £10,000 a year until the end of the patent's term in 1823.

A new type of problem now faced the patentees that not only made the royalties hard to collect but also reduced the price of the lace they produced—the overproduction that both Arkwright and Watt had feared in their own fields. Patentees and licensees came together to establish a “mart” to maintain prices: Lacy (now operating independently) was restricted to 127 frames, Heathcoat to 147, and “all parties were to use their efforts to prevent further machines being constructed.” By 1820, however, there were 1,008 licensed (and many unlicensed) frames in operation, and the price of lace was only one-fifth of what it had been three years earlier. Some licensees were even proposing higher royalties as a way of limiting output, but the situation was saved by a cyclical upturn in the market. In the long run Heathcoat, like Arkwright, was to make his fortune from his manufacturing enterprise.⁶⁴

James Kay was another manufacturer who attempted to combine direct exploitation of his invention with licensing his competitors. He fell foul of the determination of the largest firm in the linen industry, Marshalls of Leeds, not to pay anything for the right to use his wet-spinning process. They had the financial resources and the incentive of size to mount an effective defense and counterattack in the courts.⁶⁵ James Neilson seems to have been able to secure an estimated £40,000 a year in royalties by charging ironmasters the not unreasonable sum of one shilling per ton to enjoy the benefit of his hot blast, though only at the cost (allegedly £100,000) of extensive litigation.⁶⁶ John Farey thought it an “everyday occurrence” for patentees to have to fight off vexatious legal proceedings designed “merely to wear out the purse of the patentee.” To achieve this, the majority of the trade would subscribe to a “stock purse,” turning the screw further by limiting to the subscribers any compromise licensing agreement they forced on the patentee. It was some firms' standard

⁶⁴Chapman, ed., pp. 209–10, 245–50, 263–64; Dutton (n. 2 above), pp. 193–94.

⁶⁵British Patent no. 5,226 (1825); Dutton, pp. 189–92.

⁶⁶British Patent no. 5,701 (1828); BPP 1851 (n. 26 above), XVIII, p. 700; *Report of the Commissioners . . . the Working of the Law Relating to Letters Patent for Inventions*, BPP 1864, XXIX, p. 394.

practice to reject all patentees' initial offers of licenses, for fear of attesting the patent's validity by their acceptance, and to infringe with the intention of overturning the patent or reducing the cost of licenses by such tactics.⁶⁷ The most successful licensing operation under the unreformed British patent system appears to have been that conducted by E. C. Howard, which "brought in £40,000 a year during the whole time of its existence." He had taken the precaution of allocating two-fifths of the royalties as shares to the sugar refiners who were also his licensees.⁶⁸

It was not until the patent system was put on a surer footing that any patentee enjoyed major success in combining direct exploitation of an invention and its licensing. A sea change in the attitude of judges and juries toward patentees during the second quarter of the 19th century and the consequent accumulation of favorable precedents made litigation much less of a lottery than it had been. Legislation in 1835 allowed patentees to disclaim parts of their specification that weakened their case at law, without the expense of obtaining a new patent. And the Patent Law Reform Act of 1852 gave them the benefit of cheaper patent protection, combined with the opportunity to check that there was no earlier patent whose existence would vitiate their expense. Fortified by these developments, S. C. Lister notoriously made a fortune from patenting, working, and licensing wool-combing machinery; similarly, John Crossley and Sons, rather more liberally but no less successfully, made a fortune from power looms for carpet weaving.⁶⁹ The 1864 Commission of Inquiry into the working of the 1852 law found, nonetheless, that a major criticism continued to be "protracted litigation and consequent expense" and heard how both Lister and Crossley had had to contend with it. It was also able to report, however, that "patentees are, as a rule, willing to grant licences to work their inventions on reasonable terms. To do so is obviously their interest."⁷⁰

The prejudice of English judges and juries against patentees seems to have been responsible for the failure of over two-thirds of prosecutions of infringements at common law before the 1830s.⁷¹ It stemmed largely from the assumption that patents were monopolies

⁶⁷HLRO, Minutes, July 6, 1835, p. II, ff. 21–22; June 25, 1835, ff. 5–6.

⁶⁸British Patent no. 3,607 (1812), no. 3,754 (1813); BPP 1851, XVIII, p. 700; *Hansard* 28: col. 475 (June 3, 1835).

⁶⁹For Lister, see Eric M. Sigsworth, *Black Dyke Mills: A History* (Liverpool, 1958), pp. 39–41; for Crossley, see James Neville Bartlett, "Sir Francis Crossley (1817–72)," in David J. Jeremy, ed., *Dictionary of Business Biography*, vol. 1 (London, 1984), p. 848.

⁷⁰BPP 1864, XXIX, pp. 325, 331, 395, 441.

⁷¹This fell to less than one-quarter, 1830–49; Dutton (n. 2 above), p. 78.

that hindered the diffusion of innovation and that therefore the interests of patentees and society as a whole were inevitably opposed. Arguably, by their attitude they made this a self-fulfilling assessment: the lack of support offered patentees by the courts was chiefly responsible for the defensive way in which they managed their patents. Combined with the high costs of both patenting and litigating and with other uncertainties stemming from the administration's extreme *laissez-faire* approach, this insecurity led some patentees to attempt secret working; alternatively, it reduced their bargaining powers with potential licensees and assignees, who often demanded restrictive practices (notoriously, sole licenses) as a trade-off for the risk of losing their investment.⁷² It was generally assumed that the free communication of inventions under a premium system, such as that operated by the Society of Arts, would necessarily produce a wider diffusion than where an invention was protected by a patent.⁷³ This view, however, overlooked the importance of enlightened self-interest: once an inventor had received his premium, he had no further financial interest in its dissemination, beyond that of any manufacturer.

The unreformed English patent system probably promoted some diffusion of inventions "by the back door." It assisted the industrial spy to locate the sites of inventive activity, allowed him to buy specifications of some, if not complete, accuracy, and above all encouraged him in the belief that he had little cause to fear prosecution for infringement. However, in accepting this, it is not necessary to follow Harry Dutton down a Panglossian path to conclude that "the imperfect nature of the patent system . . . perhaps approached the ideal"—that, through the modicum of protection they received, inventors were sufficiently encouraged to invent, yet were unable to retard diffusion to any great degree.⁷⁴ For one thing, transaction costs were raised by these indirect routes to diffusion; for another, as I have argued, inventors' insecurity actively inhibited diffusion.

There is less contradiction between the patentee's interests and the dissemination of an invention than is generally allowed, but it does require that a patentee should believe he has the security to profit by diffusion. Although a manufacturer might wish to retain a patented process invention for his own exclusive use, this was not his sole, or necessarily his best, option. Moreover, the interest of a machine maker should normally lie in the widest possible diffusion of any

⁷²*Ibid.*, pp. 132–42.

⁷³MacLeod (n. 8 above), pp. 190–96.

⁷⁴Dutton, p. 204.

product invention consistent with market demand. (The position of the independent inventor was more indeterminate: he might assign or license his patent in either sector.) Provided cartels could be excluded, the price of a machine or a license fee would be set by what the market could bear. Whatever the shortcomings of Watt's strategy of control may have been, it remains the case that the diffusion of the steam engine was a necessary adjunct of his business as a capital-goods producer. Although he refused to license competitors to make rival products, the sale of steam engines to users was the essence of his enterprise. The premium that he charged users during the patent's thirty-one-year term was regarded as high, and doubtless encouraged many to turn to "pirate" alternatives. Yet, unlike the licensing of competitors, where royalties might be deliberately set at a deterrent level, this premium represented the cost that Boulton and Watt believed the market would bear—the calculus of maximum price consonant with maximum supply.

The compatibility of wide diffusion "through the front door" of the patent system with financial success for the individual patentee has been consummately demonstrated by Carolyn C. Cooper's study of Thomas Blanchard's adroit management of his U.S. patents for woodworking machinery, under a system that, before 1836 at the least, was very similar to England's. Here one sees a capital-goods producer employing a more confident strategy than Watt's. Blanchard decided that the way to fortune lay in widely disseminating his inventions (the irregular turning lathe, in particular), not only by manufacturing them for sale himself, but also through extensive licensing and assignment of rights. Although much of Blanchard's success lay in securing extensions of his patents beyond the single term of fourteen years (as did Watt's), and part of it in his continuing to manage the patents once licensed (which did involve some restrictive practices), the lathe was extensively and rapidly diffused throughout the United States. Cooper emphasizes the positive role played by repeated litigation and remarks that this was not a defect, but "a normal and inevitable feature of the system."⁷⁵

This presupposes, I suspect, a more certain legal situation than that enjoyed by English patentees, particularly before the 1830s. American inventors might have had to face an "interference suit" before their patent was registered but, if they survived such a challenge before the appointed tribunal of three judges, they were then in a

⁷⁵Carolyn C. Cooper, "The Roles of Thomas Blanchard's Woodworking Inventions in 19th-Century American Manufacturing Technology" (Ph.D. diss., Yale University, 1985), pp. 40–66, 358–67.

stronger position to enforce their rights. In order to achieve a similar validation English patentees had to risk the expense, delays, hostility, and uncertainties of a common-law court. Again, if a U.S. patentee began to worry about the sufficiency of his specification, it was not too costly to withdraw his patent, apply for a new one, and redraft the specification.⁷⁶ An English patentee had to find at least £100 to take the same course; only after 1835 was he allowed to disclaim a part of the specification. The U.S. system was not without its problems, particularly in the years between the death of William Thornton in 1828 and 1836, when the new act introduced substantial reforms. Thornton had struggled for twenty-six years with underfunding and shortage of staff, which led to delays, incomplete record keeping, and neglect of the patent model collection, and with the lack of administrative guidelines that promoted discord and idiosyncratic interpretations of the law. Without his experienced direction, the office's disarray became manifest.⁷⁷ The absence of examination combined with the relatively low cost of a patent promoted many disputes over priority, leading to "extensive litigation and widespread discontent with the process."⁷⁸ Nonetheless, at least there was a specialized Patent Office in existence and it offered American inventors a more accessible and cheaper service than that available in Britain. It is also worth remarking how concern over the system's inadequacies led to the enactment of reforming legislation, with the commitment of substantial public funds, at a time when similar complaints in Britain produced only abortive debates.

Before 1852, certainly before 1830, financial success usually came from directly working, not licensing or assigning, an English patent: the high probability of its being revoked, if challenged, generally reduced its market value. Comparison of the relative success of both Arkwright's and Heathcoat's respective manufacturing and licensing operations makes the point sharply. No doubt, Blanchard's strategy was simplified by his position—in the capital-goods sector, with a patent covering a product invention. One should not, therefore, attribute too much to the differences between the U.S. and English patent systems. Blanchard's case is not directly comparable with such

⁷⁶There were some risks attached to this; see, e.g., Jeremy (n. 32 above), p. 185.

⁷⁷Preston (n. 45 above), pp. 334–39, 346–53. For post-1836 developments, see Robert C. Post, " 'Liberalizers' versus 'Scientific Men' in the Antebellum Patent Office," *Technology and Culture* 17 (1976): 24–54.

⁷⁸Sokoloff (n. 27 above), p. 818; Preston, pp. 343–44; and, for an unfavorable comparison with England's equity jurisdiction, Peter A. Browne, "Mechanical Jurisprudence: On the Law of Patents," *The Franklin Journal and American Mechanics' Magazine* 4 (1827): 31–33.

as Arkwright's or Heathcoat's. We need more closely observed case studies (for which the materials are ample in the United States, if not in Britain).⁷⁹ In the manufacturing sector, we might note the similarly intrepid way in which the Boston Manufacturing Company managed its U.S. patents for textile machinery in the 1820s, combining its employment in manufacturing with its production for sale and attendant licensing on a grand scale. The company lost seven out of eight patents during infringement cases in 1828–29, making no attempt to restore them, but David Jeremy comments nonetheless that “the evidence supports the view that patents were an important method of securing infant-industry protection.”⁸⁰

Conclusion

It appears that most worthwhile inventions could escape through either the back or the front door of almost any patent system and be diffused—with varying costs and degrees of efficiency. Whether, however, their inventors would reap the benefits of that escape and therefore be encouraged to assist it, depended heavily on the configuration of the particular patent system, as structured by legislation and administrative practice and as adjudicated in the law courts. In other words, it depended on the sort of opportunities it offered them to manage their intellectual property (as Cooper has aptly described it). The 18th-century French example offers a model of state intervention to promote innovation that had been politically unacceptable in England since the Civil War and that itself was a source of conflict and internecine struggle, rejected by the revolutionary Assembly in 1791. If the U.S. patent system, created in 1790, was more supportive of innovation and its dissemination than the English system, this seemingly owed nothing to positive strategy. It was, if anything, the fortuitous outcome of a much easier access to patents, particularly with regard to cost, and to litigation; we may also speculate that public opinion was more favorable to inventors than it was in contemporary England.⁸¹ The unreformed English system (and its Scottish and Irish equivalents) placed massive hurdles in the way of inventors seeking patent protection and the profitable management of their intellectual property. That England overcame the handicap of its antiquated and inefficient patent system may have owed not a little to the emergence

⁷⁹Carolyn C. Cooper, “Thomas Blanchard's Woodworking Machines: Tracking 19th-Century Technological Diffusion,” *IA: The Journal of the Society for Industrial Archeology* 13 (1987): 41–54.

⁸⁰Jeremy, pp. 185–86.

⁸¹Eugene S. Ferguson, “The American-ness of American Technology,” *Technology and Culture* 20 (1979): 3–24.

of a specialist capital-goods sector, whose patent-management strategy was necessarily different from that of its clients. It was not that machine makers made many significant inventions themselves but rather that, when entrusted with the production of a newly invented machine, they had a vested interest in its diffusion.⁸² To reach firmer conclusions concerning actual outcomes we need more studies that follow inventors through from merely patenting into the wider industrial arena, to discover precisely how they sought to profit from their inventions.

⁸²This point is developed at length in my "Capital Goods Industries and Technological Innovation in Britain, 1780–1880" (report on E.S.R.C. grant no. AA/23/32/0032, 1989), to be published as "Strategies for Innovation: The Diffusion of New Technology in Nineteenth-Century British Industry," *Economic History Review*, forthcoming.